

Title: Quantum Decoherence in Superconducting Qubit Arrays Under Correlated Noise

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Abstract

Correlated noise sources present a major obstacle to scaling superconducting quantum processors. This paper models the effect of spatially correlated two-level-system defects on decoherence rates across arrays of transmon qubits and proposes a calibration protocol to mitigate their impact.

1. Introduction

As qubit counts increase, noise that was previously treated as independent across qubits becomes increasingly correlated due to shared substrate defects and control line crosstalk. Understanding this correlation structure is essential for designing effective error correction codes.

2. Theoretical Model

We extend the standard spin-boson model to include a shared bath of two-level-system defects coupled to multiple qubits with distance-dependent coupling strength. Master equation simulations are performed for arrays ranging from four to sixty-four qubits.

3. Results

Simulations reveal that correlated dephasing reduces the effective distance of surface code error correction by up to thirty percent compared to predictions under an independent noise assumption. Qubits separated by less than 200 micrometers on the chip exhibit dephasing correlation coefficients exceeding 0.4.

4. Mitigation Strategy

We propose a dynamical decoupling sequence tuned to the dominant correlated noise frequency, which our simulations show recovers approximately two-thirds of the lost error correction threshold margin.

5. Conclusion

Correlated noise must be explicitly modeled when designing near-term quantum error correction schemes. Chip layout choices that increase inter-qubit spacing may trade off connectivity for improved noise independence.